

Trainings ▾

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

WARNING

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

To reduce the possibility of personal injury, avoid direct contact of hot oil with your skin.

WARNING

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.

- Close the fuel supply valve. See equipment manufacturer service information.
- Disconnect the batteries. See equipment manufacturer service information.

- Drain the cooling system. Refer to Procedure 008-018 in Section 8.
(/qs3/pubsys2/xml/en/procedures/100/100-008-018.html)
- Use the following procedure for C Series diesel engines. Remove the cylinder head. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html)
- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html)
- Drain the lubricating oil. Refer to Procedure 007-037 in Section 7.
(/qs3/pubsys2/xml/en/procedures/100/100-007-037.html)
- Remove the lubricating oil pan. Refer to Procedure 007-025 in Section 7.
(/qs3/pubsys2/xml/en/procedures/100/100-007-025.html)
- Remove the piston and connecting rod assemblies. Refer to Procedure 001-043 in Section 1.
(/qs3/pubsys2/xml/en/procedures/100/100-001-043.html)

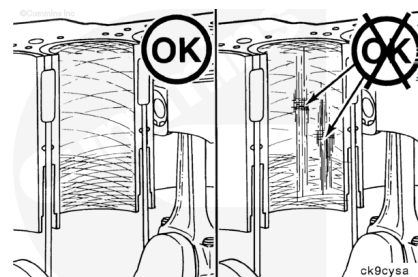
Initial Check

Note : Before removing the cylinder liners, inspect the condition to determine reusability.

Inspect the inside of the liner for cracking, scuffing, or scoring.

Inspect the inside for vertical scratches deep enough to be felt with a fingernail.

Note : If a fingernail catches in a scratch, the liner must be replaced.



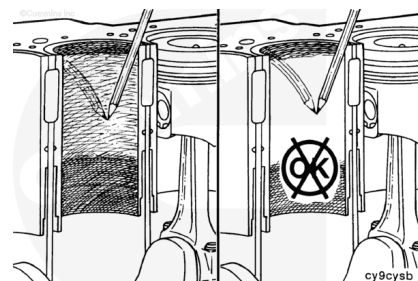
Inspect the inside of the liner bore for polishing.

Note : A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

Note : A heavy polish produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.

Do **not** reuse liners that have heavy polish or any indications of particle embedment from failure.

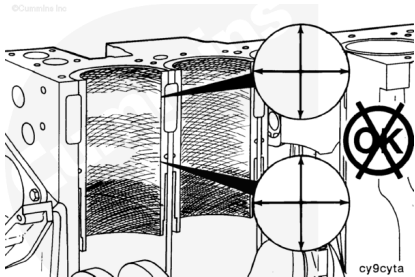
Inspect the liners for a ridge near the top of the liner. Severely worn liners will have a ridge near the top of the liner bore.



If the wear pattern indicates the liner bore is **not** straight or **not** round, use a dial bore gauge to measure the liner inside diameter in four places 90 degrees apart at the top and bottom

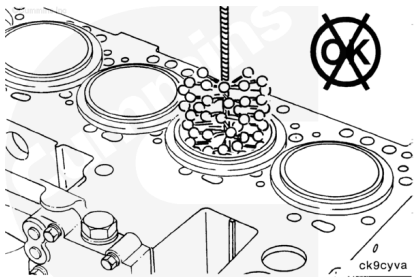
of the piston travel area.

Cylinder Liner Dimensions				
	mm		in	
Out of Round		0.04	MAX	0.002
Taper		0.04	MAX	0.002
Bore Diameter		114.04	MAX	4.490



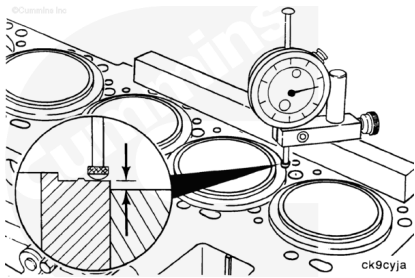
Note : If the liner bore is not acceptable for reuse, it must be replaced. Damaged liners can not be honed or deglazed. This will destroy the crosshatch pattern needed for oil control.

Note : Do not remove or disturb the liner seals for liners that pass the inspection criteria.



Cylinder liner protrusion is the distance the liner protrudes above the block face. Before removing the liners, check the protrusion at four places 90 degrees apart. Use the liner protrusion gauge, Part Number 3823495, in the unclamped state.

Cylinder Liner Protrusion			
	mm		in
New Block and Liners	0.025	MIN	0.001
	0.122	MAX	0.005
Used Block and Liners	0.000	MIN	0.000
	0.122	MAX	0.005
New Block Used Liners	0.025	MIN	0.001
	0.122	MAX	0.005
Used Block New Liners	0.000	MIN	0.000
	0.122	MAX	0.005

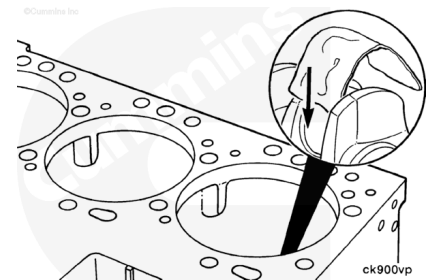


The maximum allowable difference between adjacent cylinders is 0.051mm [0.002 in].

Note : If the liner was installed correctly and the liner protrusion is out-of-limits, it will be necessary to machine the counterbore and add shims to restore to the original specifications.

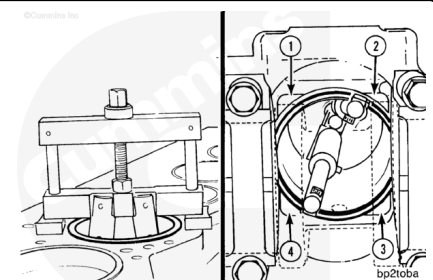
Remove

Note : Use clean shop rags to cover the crankshaft to prevent debris from falling into the main journal area or into the connecting rod journal oil drilling.



⚠ CAUTION ⚠

The liner puller must be installed and used as described to avoid damage to the cylinder block. The puller must not contact the block casting at points (1), (2), (3), or (4).

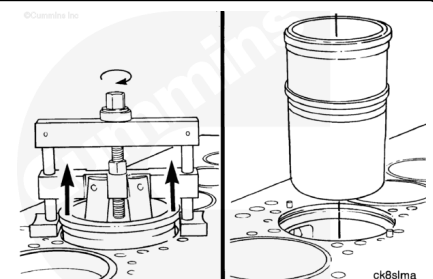


Insert the cylinder liner puller, Part Number 3376015, into the top of the cylinder block.

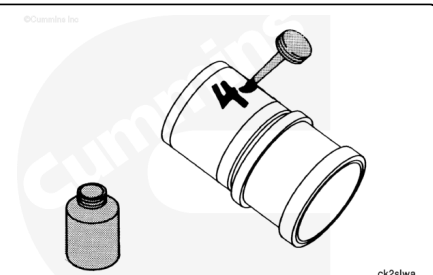
Note : The liner puller must be centered on the top of the cylinder block.

Turn the puller jackscrew **clockwise** to loosen the liner from the cylinder block.

Use both hands to remove the liner.



Mark the cylinder number on each liner.

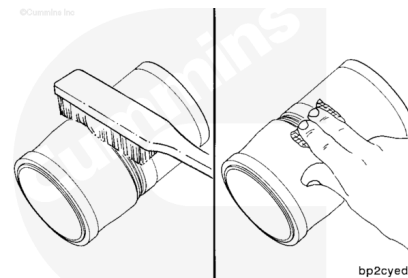


Clean and Inspect for Reuse

⚠ CAUTION ⚠

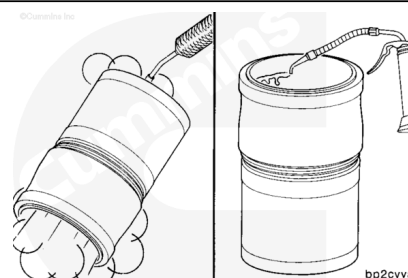
**Do not use any abrasives in the ring travel area of the liner.
The liner can be damaged.**

Use a soft wire brush or a fine fibrous abrasive pad such as Scotch-Brite® 7448, Part Number 3823258, to clean the flange seating area.



⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use solvent or steam to clean the cylinder liners.

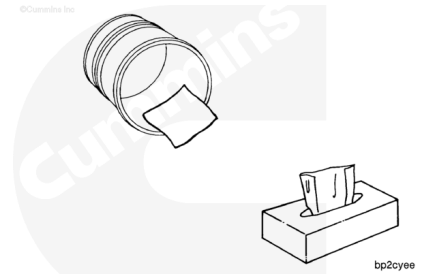
Dry the cylinder liners with compressed air.

Use clean engine oil to lubricate the inside of the liners.

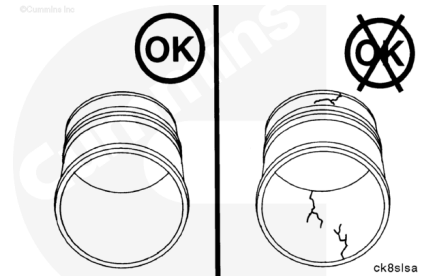
Allow the oil to soak in the liner for 5 to 10 minutes.

Note : Use lint-free paper towels to wipe the oil from the inside of the liners.

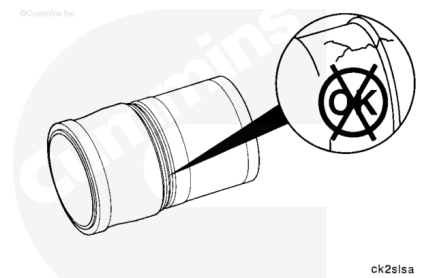
Continue to lubricate the inside surfaces of the liners and wipe clean until the paper towel shows no gray or black residue.



Inspect the liners for cracks on the inside and outside surfaces.



Inspect for cracks under the flange.

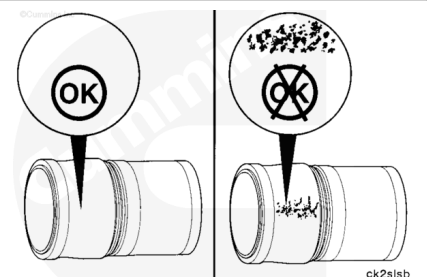


Inspect the outside and seat area for excessive corrosion or pitting.

Note : Pits must not be more than 1.6 mm [0.06 in] deep.

Replace the liner if the pits are too deep or if the corrosion can **not** be removed with Scotch-Brite® 7448, Part Number 3823258.

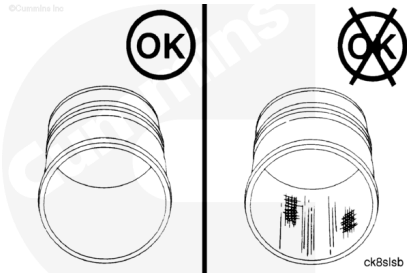
Note : Cylinder liner pitting is caused by poor coolant composition. Refer to the Operation and Maintenance Manual, GTA8.3 CM558/CM2358, Bulletin 4021640, for coolant specifications.



Inspect the inside surface for vertical scratches deep enough to be felt with a fingernail.

Note : If a fingernail catches in the scratch, the liner must be replaced.

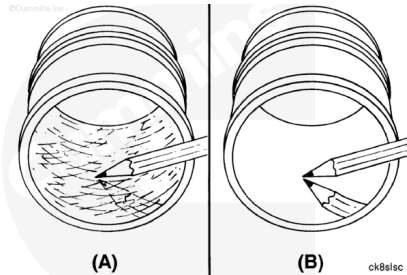
Inspect the inside surface for scuffing or scoring.



Inspect the inside surface for liner bore polishing.

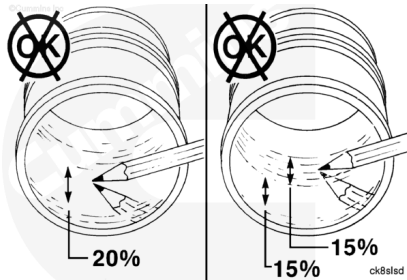
Note : A moderate polish produces a bright mirror finish in the worn area with traces of the original hone marks or an indication of an etch pattern.

Note : A heavy polish produces a bright mirror finish in the worn area with no traces of hone marks or an etch pattern.



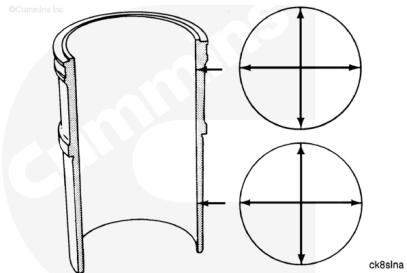
Replace the liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- Both moderate and heavy polish over 30 percent of the piston ring travel area and one half (15 percent) is heavy polish.



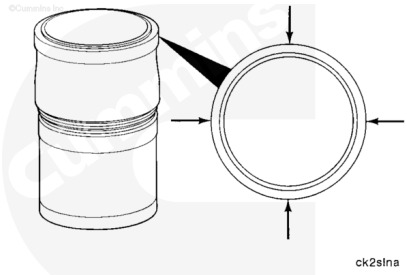
Measure the liner inside diameter with dial bore gauge, Part Number 3375072, in four places 90 degrees apart at the top and bottom of the piston travel area.

Cylinder Liner Inside Diameter		
mm		in
114.00	MIN	4.488
114.04	MAX	4.490



Measure the outside diameter of the liner top press fit area.

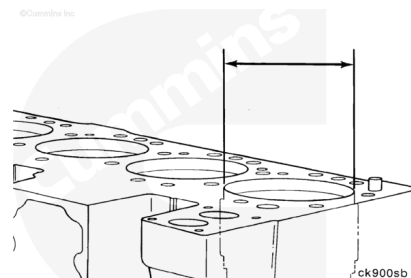
Cylinder Liner Outside Diameter (Top Press Fit)		
mm		in
130.938	MIN	5.1550
130.958	MAX	5.1558



Measure the cylinder liner bore in the block.

Cylinder Liner Bore in Block (Press-Fit Bore)

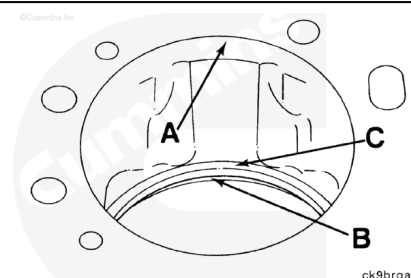
mm		in
130.90	MIN	5.154
130.95	MAX	5.156



Install

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

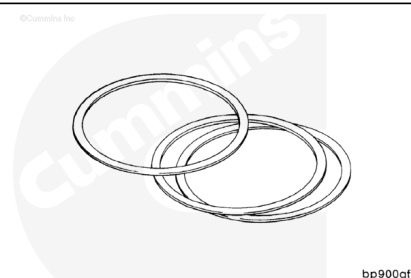
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Clean all deposits and debris from the sealing surfaces (A), (B), and (C). Use Scotch-Brite® 7448, Part Number 3823258, and cleaning solvent to polish the surfaces. Due to the critical machined tolerances, care **must** be taken **not** to remove any additional material.

Use the following procedure if surface C has signs of cracking, fretting, or pitting for inspection, reuse, and repair guidelines. 001-026 in Section 1.

Cylinder liner shims are available as follows:

Measurements			
		mm	in
Cylinder Liner Shim Thickness - By Part Number	3924445	0.25	0.010

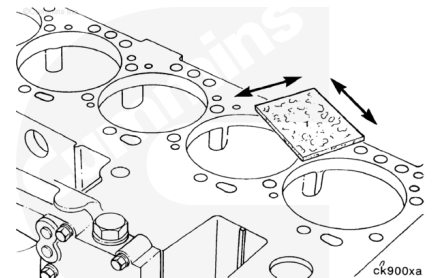


Measurements			
		mm	in
	3924446	0.38	0.015
	3924447	0.51	0.020
	3924448	0.76	0.030
	3924449	1.00	0.040

Use a maximum of **only** one shim.

⚠ WARNING ⚠

When using solvents, acid, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



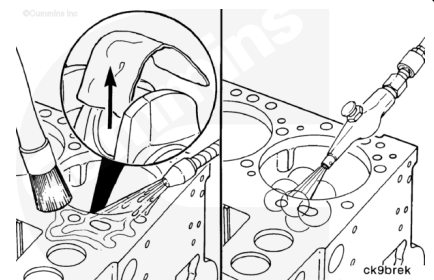
⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Clean the combustion deck with a gasket scraper or Scotch-Brite® 7448, Part Number 3823258, and solvent.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause bodily injury.



⚠ WARNING ⚠

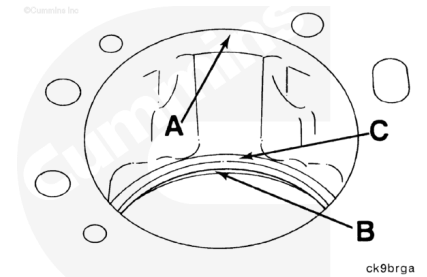
Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

Thoroughly flush the block with mineral spirits or cleaning solvent.

Remove the shop rags and clean the crankshaft with a cleaning solvent.

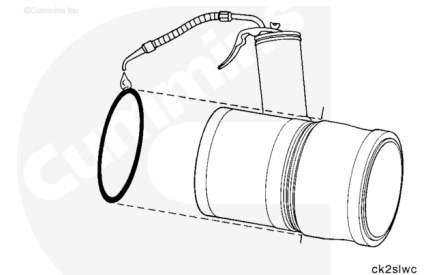
Blow the cylinder bores and crankshaft dry, and wipe clean with a lint-free cloth.

Lubricate surfaces (A) and (B) with clean engine oil.

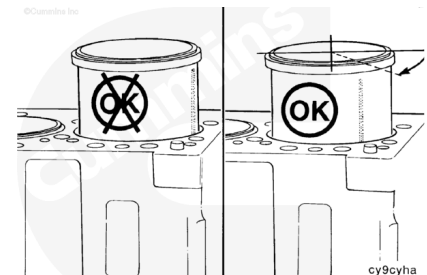


Use clean engine oil to coat the liner o-ring seals.

Install the new lubricated o-ring seals on the cylinder liners.

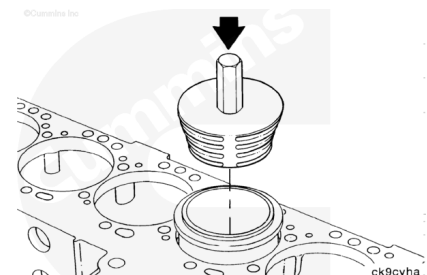


Note : When reusing liners, install them in the same cylinder from where they were removed, and rotate the liner 45 degrees (1/8 of a turn) from their original position. When correctly installed, any liner pitting can be positioned as illustrated so the pitted surface is rotated away from the location where pitting occurs.



Install the cylinder liners into the bore of the cylinder block, use cylinder liner driver, Part Number ST-1229, and a leather mallet.

Note : If the liner does not rest on the cylinder block counterbore seat, remove the liner. Inspect the counterbore seat and liner for nicks, burrs, or dirt. Install the liner again.

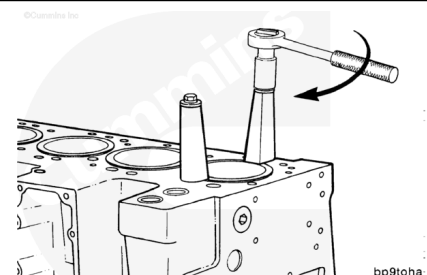


Use two cylinder head capscrews and position the two liner clamps, Part Number 3822503, as illustrated.

Tighten the capscrews.

Torque Value: 68 n·m [50 ft-lb]

Remove the clamps and repeat this procedure until all liners have been clamped and released.

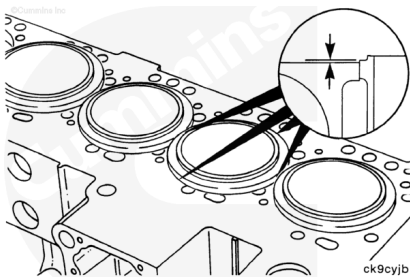


Liner protrusion is the distance the liner protrudes above the block face.

Measure the liner protrusion with the liner protrusion gauge, Part Number 3823495, at four points 90 degrees apart.

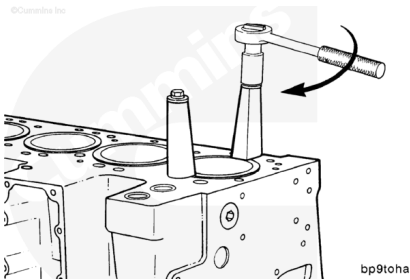
Cylinder Liner Protrusion		
mm		in
0.025	MIN	0.001
0.122	MAX	0.005

The maximum allowable difference between adjacent cylinders is 0.051mm [0.002 in].

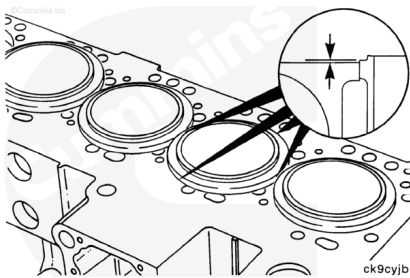


Note : If the cylinder liner protrusion varies more than 0.025 mm [0.001 in] in 180 degrees:

- Install and tighten the cylinder liner clamps again.

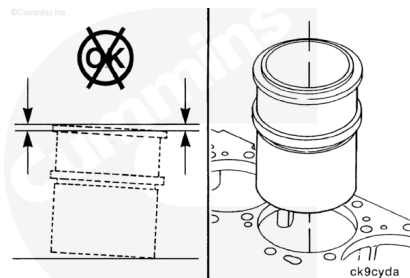


- Check the cylinder liner protrusion again.

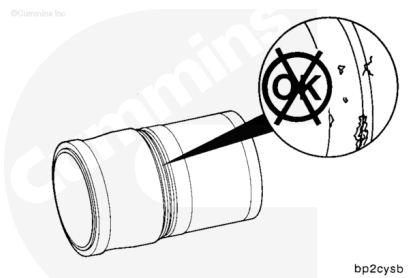


Note : If the protrusion still varies more than 0.025 mm [0.001 in]:

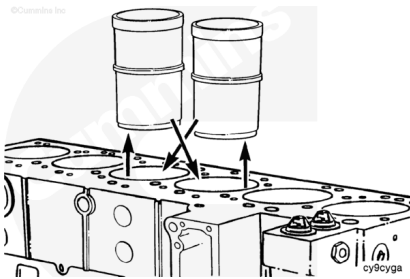
- Remove the cylinder liner.



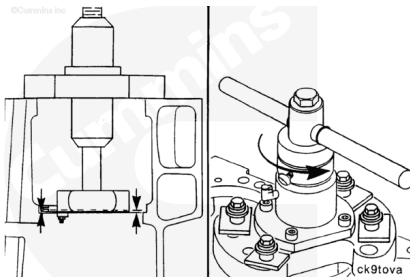
- Inspect the cylinder liner sealing edge for burrs, dirt, or damage.
- Replace the cylinder liner if it is damaged.
- Install the cylinder liner again.
- Check the cylinder liner protrusion.



Service Tip: If the out-of-limit condition is minimal, tolerance stackup will allow the protrusion limits to be obtained by installing other new liners in the out-of-limit bore.

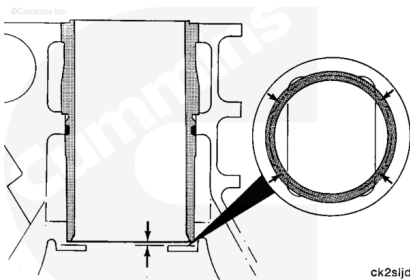


Note : Use the following procedure if the liner protrusion still does not meet the specifications Refer to Procedure 001-058 in section 1 (/qs3/pubsys2/xml/en/procedures/100/100-001-058.html) to machine the cylinder block liner counterbore for shims.



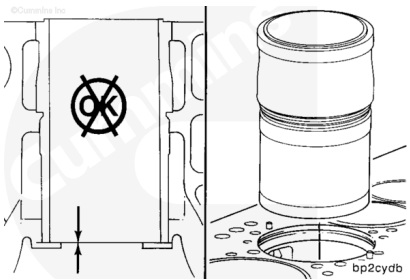
Use a feeler gauge to inspect the liner to block clearance at the four block casting points.

Cylinder-Liner-to-Block Clearance		
mm		in
0.23	MIN	0.009

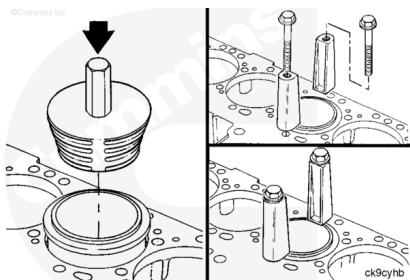


Note : If the clearance is less than 0.23 mm [0.009 in]:

- Inspect the liner and cylinder block for dirt or damage.
- Remove the liner.

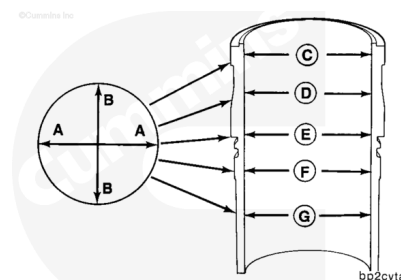


- Replace the liner if it is damaged.
- Install the liner again.



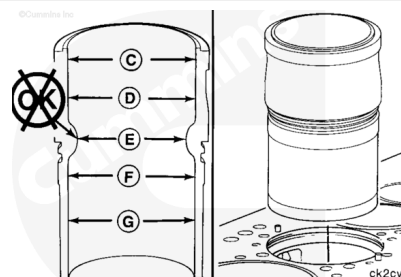
Measure the liner bore for out-of-roundness at points (C), (D), (E), (F), and (G).

Measure each point in the direction (AA) and (BB). The bore **must not** be more than 0.08-mm [0.003-in] out of round.



Note : If the bore is more than 0.08-mm [0.003-in] out of round:

- Remove the liner so the cylinder block liner bore can be measured.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

Note : Do not interchange pistons from the original cylinder location.

- Install the piston and connecting rods. Refer to Procedure 001-043 in Section 1. (</qs3/pubsys2/xml/en/procedures/100/100-001-043.html>)
- Install the lubricating oil pan and suction tube. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-025.html>)
- Use the following procedure for C Series diesel engines. Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/41/41-002-004-tr.html>)
- Use the following procedure for C8.3G, C Gas Plus, and L Gas Plus engines. Install the cylinder head. Refer to Procedure 002-004 in Section 2. (</qs3/pubsys2/xml/en/procedures/39/39-002-004-tr.html>)
- Fill the cooling system. Refer to Procedure 008-018 in Section 8. (</qs3/pubsys2/xml/en/procedures/100/100-008-018-tr.html>)
- Fill the lubricating system. Refer to Procedure 007-037 in Section 7. (</qs3/pubsys2/xml/en/procedures/100/100-007-037.html>)
- Connect the batteries. See equipment manufacturer service information.
- Open the fuel supply valve. See equipment manufacturer service information.

- Operate the engine until the coolant temperature reaches 82°C [180°F]. Check for leaks and proper operation.

Last Modified: 25-Mar-2019
